Build a complete prediction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net force, velocity change, and final position in one model.

SUCCESS CRITERIA - I can...

1. Connect force, acceleration and position consistently.
2. Show quantities and units at every stage.
3. Explain the assumptions and limits of the prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “reference frame” and “net force” to explain your model or result. Sentence starters are optional.

reference frame: the origin and directions used to describe motion

net force: the vector sum of all forces on one object

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

Maximum static friction is 50 N. A 20 N push leaves a box still.
Find actual friction. A 60 N push starts sliding with 40 N kinetic friction; find net force.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

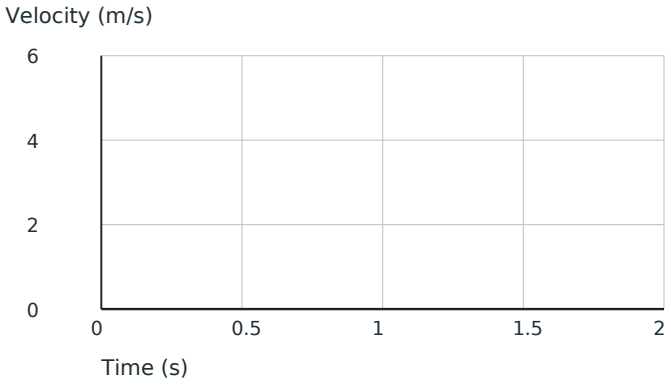
02 Worked example: explain the model

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$; $x_f = x_i + \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; use one frame. i = initial; f = final. Apply each direction separately.

Worked reasoning: A 2 kg cart starts from rest under 6 N right and 2 N left for 2 s. Net force 4 N gives $a=2$ m/s². Its displacement is $\frac{1}{2}\times2\times2^2=4$ m right.

Your turn: Draw the force diagram and the matching velocity-time triangle.
Check the 4 m using its area.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Start with forces

A 2 kg cart starts at rest. There are 6 N right and 2 N left. **Find** net force and acceleration.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.



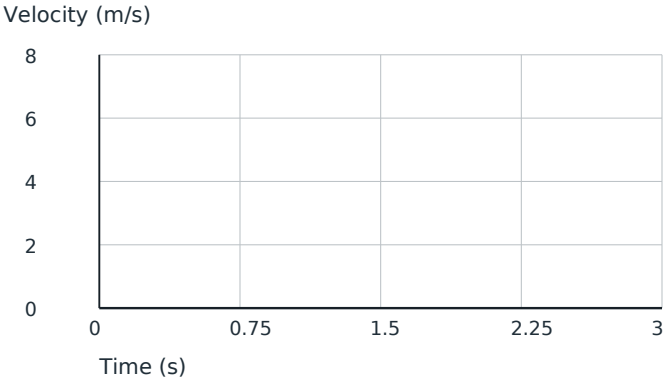
The dot represents the object. Mass 2 kg; starts at rest

Marzano check - circle: 1 2 3

04 Predict motion

The forces on the cart in question 3 stay constant for 3 s. **Draw** its velocity-time graph, then find final velocity, average velocity and displacement. If the cart starts at the 1 m mark, where does it finish?

Sentence starter: I found ... [unit]. This value means ... in this situation.

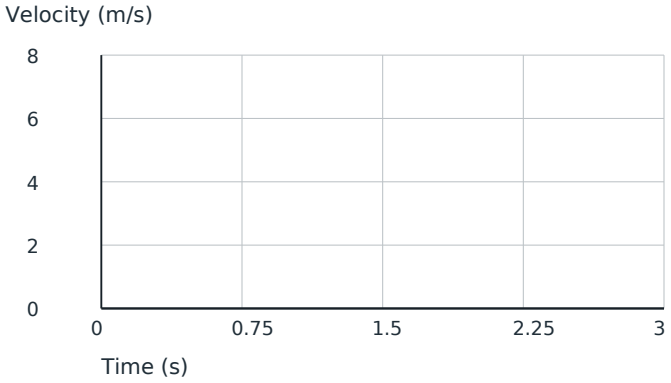


Marzano check - circle: 1 2 3

05 Explain a limit

Your prediction in question 4 assumes constant net force. **Name** a change that could break that assumption. **Draw** how the velocity-time line would differ if a larger, constant opposing friction force acted throughout.

Sentence starter: Both ... . They differ in ... because ... .



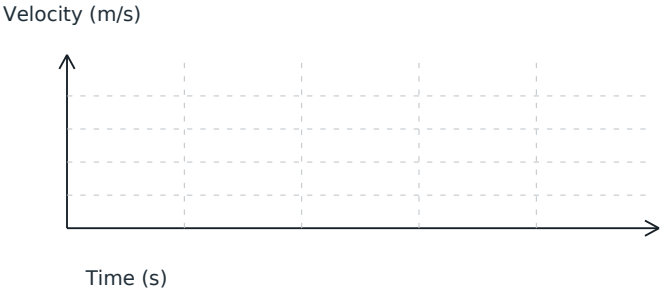
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart accelerates uniformly from rest to 14 m/s east in 5 s.
Find acceleration, average velocity and displacement.
Sketch the velocity-time model; label the given values.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

Build a complete prediction

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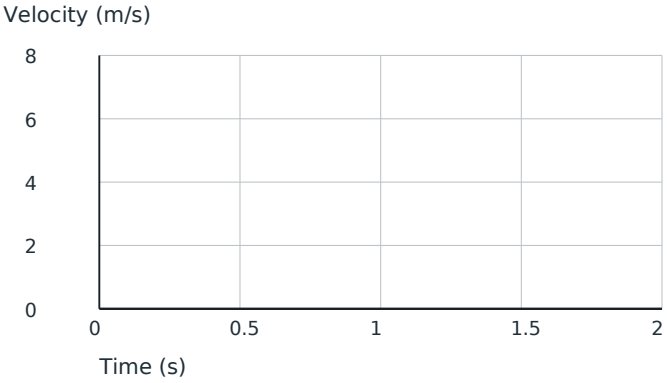
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A 3 kg rover starts from rest with a constant 9 N net force to the right for 2 s.
Find acceleration and final velocity.
Draw a graph that checks your result, and state one assumption.

Sentence starter: My claim is … . The diagram/data show …, so … .

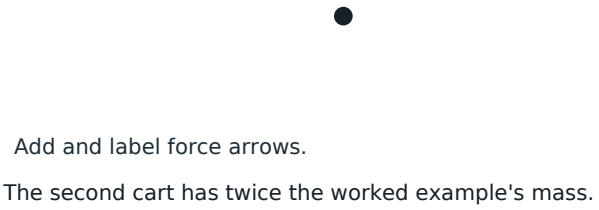


Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. A second cart must cover the same 4 m in 2 s from rest but has twice the mass. What net force is required?
Show your chain of reasoning.

Sentence starter: I notice … . This supports … because … .



Add and label force arrows.

The second cart has twice the worked example's mass.

Marzano check - circle: 1 2 3